



# TASKFLOWAI

## AI Powered Smart Task Management System

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### Python Development Internship Report

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Organization: Synent Technologies

### Acknowledgement

I sincerely thank Synent Technologies for providing this internship opportunity. This project enhanced my practical knowledge of Flask, databases, authentication, frontend development, and AI integration.

### Abstract

TaskFlowAI is an AI-powered task management web application developed using Flask, SQLite, Bootstrap, SQLAlchemy, and Google's Gemini API. It enables task management, AI planning, analytics, and productivity tracking.

### Objectives

- Develop a Flask web application
- Implement authentication
- Build CRUD operations
- Integrate Gemini AI
- Provide dashboard analytics
- Create a responsive UI

## Technology Stack

Python, Flask, SQLAlchemy, SQLite, Flask-Login, Flask-WTF, Bootstrap 5, Chart.js, HTML, CSS, JavaScript, Google Gemini API

## System Architecture

User → Flask → Authentication / Tasks / AI → SQLite → Gemini API

## Modules

Authentication  
Task Management  
Dashboard  
AI Planner  
AI Analysis  
AI Coach  
Analytics

## Features

Registration/Login  
Dashboard  
CRUD Tasks  
Search & Filters  
AI Planner  
AI Task Analysis  
Save AI Plan  
Charts  
Dark Mode

## Database Design

Users(id, username, email, password)  
Tasks(id, title, description, category, priority, due\_date, status, user\_id)

## Workflow

Login → Dashboard → Create Task → AI Analysis → Save Task → Track Progress

## Challenges

Blueprint configuration, Gemini API integration, migrations, JSON parsing, responsive UI, deployment.

## Future Scope

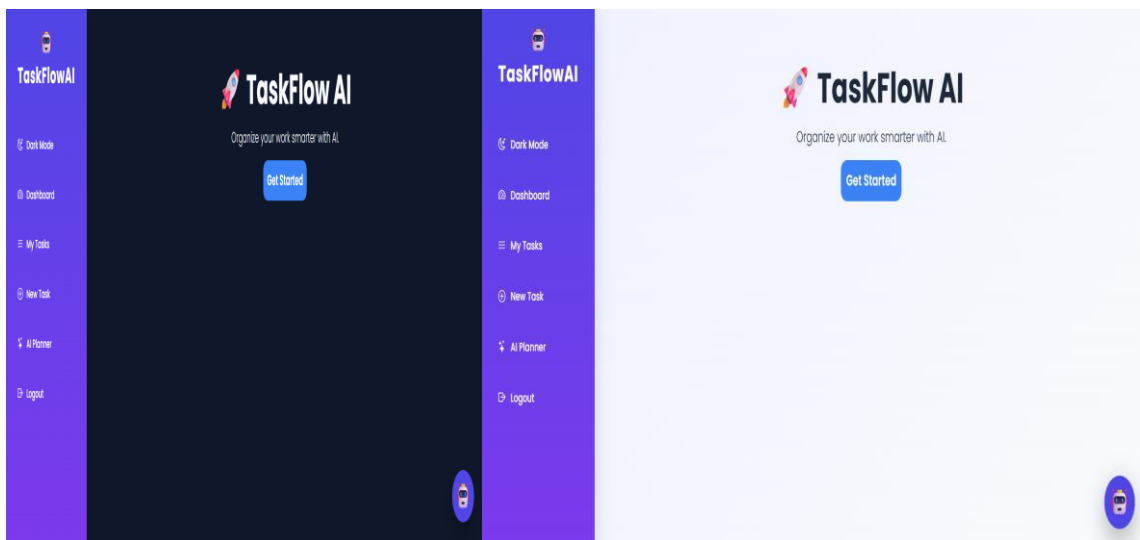
Email reminders, PostgreSQL, calendar, collaboration, mobile app, improved AI assistant.

## Conclusion

The project demonstrates full-stack Python development, database design, AI integration, and modern web application practices.

## Screenshots

Home Screen



TaskFlowAI

Dark Mode

Dashboard

My Tasks

New Task

AI Planner

Logout



Join TaskFlowAI

Create your account and start managing tasks with AI.

Username

Enter your username

Email

Enter your email

Password

Create a password

Confirm Password

Confirm your password

Register

Already have an account?  
[Login](#) >>



TaskFlowAI

Dark Mode

Dashboard

My Tasks

New Task

AI Planner

Logout



TaskFlowAI

Welcome back! Sign in to continue.

Email

Enter your email

Password

Enter your password

Login

Don't have an account?  
[Create Account](#) >>



Upcoming Deadlines

xyz

3 day(s) overdue

paper

2 day(s) overdue

Task Status

Completed

Pending

Completed

2

Categories

AI Planner

Internship

College

USA

New Task

My Tasks

AI Planner

Recent Tasks

View All →

Master the fundamentals of Flask to build and deploy a dynamic web application in 10 days.

• Day 1: Environment Setup & Routing - Install Flask in a virtual environment, understand the application context, and create basic routes and dynamic URLs. • Day 2 & 3: Templates & Static Files - Master Jinja2 templating, template inheritance, context processors, and integrating static CSS/JS assets. • Day 4: Forms & Request Handling - Implement Flask-WTF for secure form handling, CSRF protection, custom validation, and rendering flash messages. • Day 5 & 6: Database Integration - Connect Flask-SQLAlchemy, define database models, perform CRUD operations, and manage migrations using Flask-Migrate. • Day 7 & 8: User Authentication - Implement secure registration, login/logout flows, password hashing with Werkzeug, and session management using Flask-Login. • Day 9: Capstone Project Construction - Build a complete, functional CRUD application (e.g., a microblog or a task manager) integrating all previous concepts. • Day 10: Testing, Security & Deployment - Write basic unit tests, configure production environment variables, and deploy the application to Render, Fly.io, or PythonAnywhere. AI Tips: 1. Hands-on coding: Avoid passive reading; type every line of code manually to build muscle memory. 2. Embrace the debugger: Keep Flask's debug mode active and learn to read stack traces to resolve errors independently. 3. Read the Docs: Flask's official documentation is exceptionally well-written and serves as the best reference. 4. Build small: Avoid feature-creep; keep your capstone project minimal and functional before trying to polish the UI.

AI Planner

Medium

Pending

nbjh

Internship

Low

✓ Completed

Due: 2026-07-01

xyz

Internship

Medium

Pending

paper

College

High

Pending

My Tasks

New Task

Search Task

All Categories

All Categories

College

Internship

Personal

AI Planner

All Priorities

All Status

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Completed

Internship

Low

2026-07-01

**+ Create New Task**

Task Title

### Description

Category

Internship

Priority

Due Date

### Create Task

**+ Create New Task**

Task Title

### Description

Category

Internship

Priority

Due Date

### Create Task



# TaskFlowAI

 Dark Mode

 Dashboard

 My Tasks

 New Task

 AI Planner

 Logout